WASTEWISE SCAN · FEASIBILITY ASSESSMENT

WasteWise Scan feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First feasibility assessment (kit 1.8 upgrade) of option A, the current design

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Option A is feasible on paper to build and run as a proof-of-concept prototype: the optics, power, timing, sunlight handling and eye safety close by calculation on stated assumptions (fixes F1 to F3 made in this session), and every part can be bought or printed and assembled with hand tools (WSC-BLD-001). The open question is performance, not buildability: whether eight bands tell PE from PP to 95 % (PR-001, rated Medium), which only a labelled item set at TRL 4 can show. Mass (9 g over), direct-sun legibility and the price of the SWIR parts are the other weak points. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design: optical principle	Discrete-band NIR reflectance with one InGaAs photodiode identifies resins in prior work; signal-to-noise ratio of 640 or more in shade with the four-cycle scan of fix F2; the window crosstalk glass path closed by fix F1 (wall leak to measure)	High	Item scatter decides accuracy; backed mode needs its own training data	WSC-STM-001 S1 to S7, S12; WSC-CAL-001 B, C, D
Design: classification accuracy	Paper band study 89.5 % for five resins, 93 % for PE against PP at 1.5 % item scatter; 96.7 % at 1.0 %	Medium	Approximate reference spectra, not measured items; an extended InGaAs band may be needed (WSC-DEC-001 open decision 1)	WSC-CAL-001 B4 to B7; STEM S23
Design: mass and handling	239 g, 259 g with the cap, against 250 g	Medium	9 g over; thinner display frame or head block, or counting without the cap, are the remaining levers	WSC-CAL-001 A3; PR-006
Design: display in sun	5.2:1 where the hood shades the screen, 1.6:1 otherwise	Medium	Depends on how the user holds it; colour and icon cues stay primary	WSC-CAL-001 I; PR-007
Materials	PETG and TPU prints, borosilicate windows, PTFE reference disc; carbon-black PETG for the head block (fix F1)	High	Printed PETG strength along layers (strap margin 1.45 at the peak drop load after fix F3); filament opacity in NIR to confirm	WSC-CAL-001 J4, C8; WSC-DEC-001 confirm items 10, 11
Manufacturing and assembly	FDM printing, heat-set inserts, through-hole soldering on perfboard, one 8.4 mm cored hole in glass; 21 components in 13 steps	High	Core-drilling glass needs care and safety glasses; SWIR LED pockets need reaming	WSC-BLD-001; DWG-101 to DWG-109
Power and controls	One protected 18650 cell, 21.9 h at 4 scans per minute; ESP32-S3 board with its charger; ADS1220-class converter	High	Charging through the slide switch to confirm on the board; hardware LED on-time limit to design	WSC-CAL-001 F; WSC-DEC-001 confirm item 3
Supply of parts	Display board, cell, switch and screws are common; SWIR LEDs and InGaAs photodiodes come from specialist distributors	Medium	SWIR LEDs about USD 8 to 15 each (indicative); a named 1 mm InGaAs photodiode lists at EUR 104 against USD 30 in the BOM	BOM lines 3, 4; WSC-MKT-001
Software	Classifier (LDA or small tree) on the ESP32-S3; local price table; 64-byte records and CSV export	High	Firmware is TRL 4 work beyond a labelled sketch	WSC-DDR-001 D5; WSC-DDR-002
Regulatory	Eye safety exempt by estimate; cell and radio approvals to assess in India	Medium	WPC ETA of the board's module; BIS registration of the cell	WSC-REG-001 R1 to R6